

CONSULTING AND ANALYTICS CLUB, IIT GUWAHATI

Time Series Analysis

Capstone Project 2026 — Detailed Report

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Platform: StockGro | Virtual Capital: Rs. 10,00,000 | Trading Window: 14–15 May 2026

1. Methodology and Models Used

This project applies time series forecasting to construct and execute a data-driven virtual equity portfolio of Rs. 10,00,000 on StockGro. Two datasets were used to isolate model evaluation from live forecasting: Dataset A (Train: 2021–Dec 2025, Test: Jan–Mar 2026) for unbiased MAPE and accuracy metrics; Dataset B (2021–13 May 2026, full retrain) for actual 5-day price forecasts used in trading. Three models were built per stock — ARIMA (AIC-based order selection), Prophet (weekly and yearly seasonality), and LSTM (60-day lookback, two-layer, dropout 0.2, autoregressive 5-day forecast loop). Forecasts were averaged into an ensemble used for portfolio construction.

2. Stock Selection Rationale (Task 1)

Stock	Ticker	Sector	Rationale
HDFC Bank	HDFCBANK.NS	Banking	Largest private sector bank; stable post-merger uptrend; moderate volatility; strong mean-reverting properties suited to ARIMA.
Infosys	INFY.NS	IT	Strong sector momentum; quarterly earnings-driven dynamics create forecast opportunity; cleaner price series than TCS.
Sun Pharma	SUNPHARMA.NS	Pharma	Low market correlation; defensive position; volatility driven by FDA decisions rather than macro — provides diversification.
HUL	HINDUNILVR.NS	FMCG	Lowest annualised volatility (20.4%); consistent uptrend; strong seasonal component visible in STL — ideal portfolio anchor.
Maruti Suzuki	MARUTI.NS	Auto	Market leader in passenger vehicles; strong revenue visibility; high-volatility profile useful for GARCH modelling and stress-testing.

Stock Universe — Close Prices, January 2021 to May 2026



Figure 1: Historical Close Prices — All 5 Stocks (Jan 2021 – May 2026)

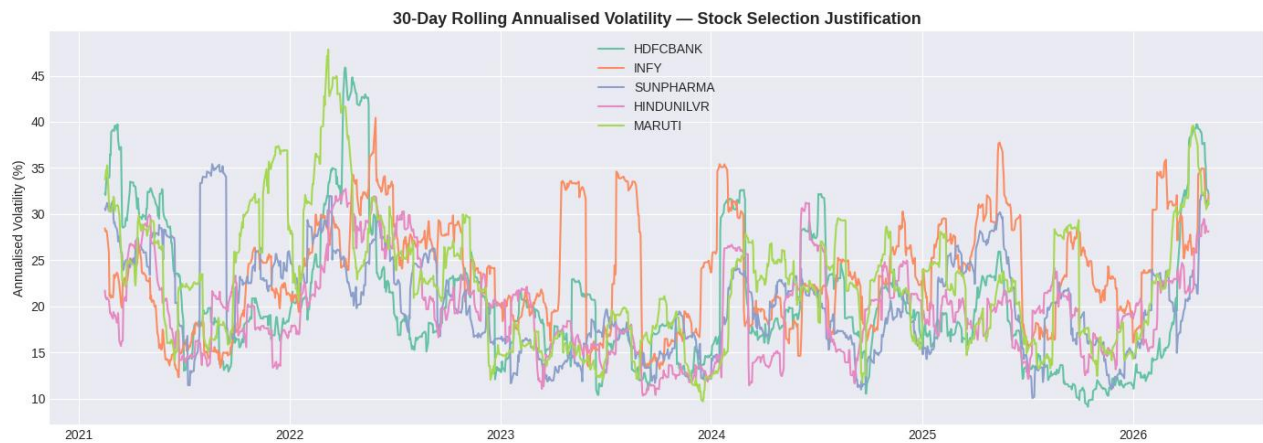


Figure 2: 30-Day Rolling Annualised Volatility — Stock Selection Justification

3. Data Preprocessing (Task 2)

Daily adjusted close data was fetched via yfinance (1,236–1,324 trading days per stock). Missing values filled via forward-fill. Dataset A: 1,236 training rows, 60 test rows per stock. ADF test confirmed all stocks non-stationary in levels ($p > 0.05$) but stationary after first differencing ($p = 0.000$), confirming $d = 1$ for ARIMA. MinMaxScaler was fitted exclusively on training data for LSTM to prevent data leakage — the full series was then transformed using training-derived parameters only.

Stock	ADF (Levels)	p-value	Result (Levels)	Result (Differenced)
HDFCBANK	-1.394	0.585	Non-stationary	Stationary ($p=0.000$)
INFY	-2.407	0.140	Non-stationary	Stationary ($p=0.000$)
SUNPHARMA	-0.851	0.804	Non-stationary	Stationary ($p=0.000$)
HINDUNILVR	-2.890	0.047	Borderline (5%)	Stationary ($p=0.000$)
MARUTI	0.503	0.985	Non-stationary	Stationary ($p=0.000$)

4. Forecast Results with Confidence Intervals (Task 3)

ARIMA orders selected by AIC: HDFCBANK (3,1,3), INFY (0,1,0), SUNPHARMA (3,1,3), HINDUNILVR (0,1,0), MARUTI (2,1,0). LSTM used a 60-day lookback with two stacked layers (64→32 units) and dropout 0.2; 5-day forecasts generated via an autoregressive loop. Prophet provided native 95% confidence intervals. After evaluation on Dataset A, all models were retrained on Dataset B for final forecasts.

ARIMA Model — Actual vs Predicted Prices, Evaluation Dataset



Figure 3: ARIMA — Actual vs Predicted (Test: Jan–Mar 2026)

Prophet Model — Actual vs Predicted with Confidence Intervals, Evaluation Dataset



Figure 4: Prophet — Actual vs Predicted with 95% CI (Test: Jan–Mar 2026)

LSTM Model — Actual vs Predicted Prices, Evaluation Dataset



Figure 5: LSTM — Actual vs Predicted (Test: Jan–Mar 2026)

Ensemble 5-Day Forecasts (14–20 May 2026)

Stock	Day 1 (Rs.)	Day 2 (Rs.)	Day 3 (Rs.)	Day 4 (Rs.)	Day 5 (Rs.)
HDFCBANK	786.77	783.96	783.24	784.19	784.74
INFY	1,164.35	1,163.90	1,163.82	1,165.04	1,165.65
SUNPHARMA	1,786.84	1,786.84	1,788.71	1,788.96	1,787.16
HINDUNILVR	2,276.57	2,275.05	2,276.47	2,281.30	2,283.62
MARUTI	13,972.33	13,973.88	13,975.49	13,987.63	13,995.33

Day 1 = Thu 14 May | Day 2 = Fri 15 May | Day 3 = Mon 18 May | Day 4 = Tue 19 May | Day 5 = Wed 20 May

5. Volatility and Trend Analysis (Task 4)

Stock	Sector	Ann. Vol (%)	30D Vol (%)	Skewness	Kurtosis	Trend
HDFCBANK	Banking	21.69	31.98	-0.081	4.998	Upward
INFY	IT	24.68	32.53	-0.401	4.118	Upward
SUNPHARMA	Pharma	21.11	30.75	0.482	3.558	Upward
HINDUNILVR	FMCG	20.40	27.98	-0.014	2.962	Upward
MARUTI	Auto	23.64	31.37	0.292	3.519	Upward

All stocks show kurtosis > 3 (fat tails). INFY has the highest annualised vol and strongest negative skew. HINDUNILVR is the lowest-volatility stock, confirming its role as the portfolio anchor. GARCH(1,1) Day 1 forecasts: HDFCBANK 1.62% | INFY 1.94% | SUNPHARMA 1.36% | HINDUNILVR 1.34% | MARUTI 1.73%. These directly inform the inverse-volatility weights in Strategy B.

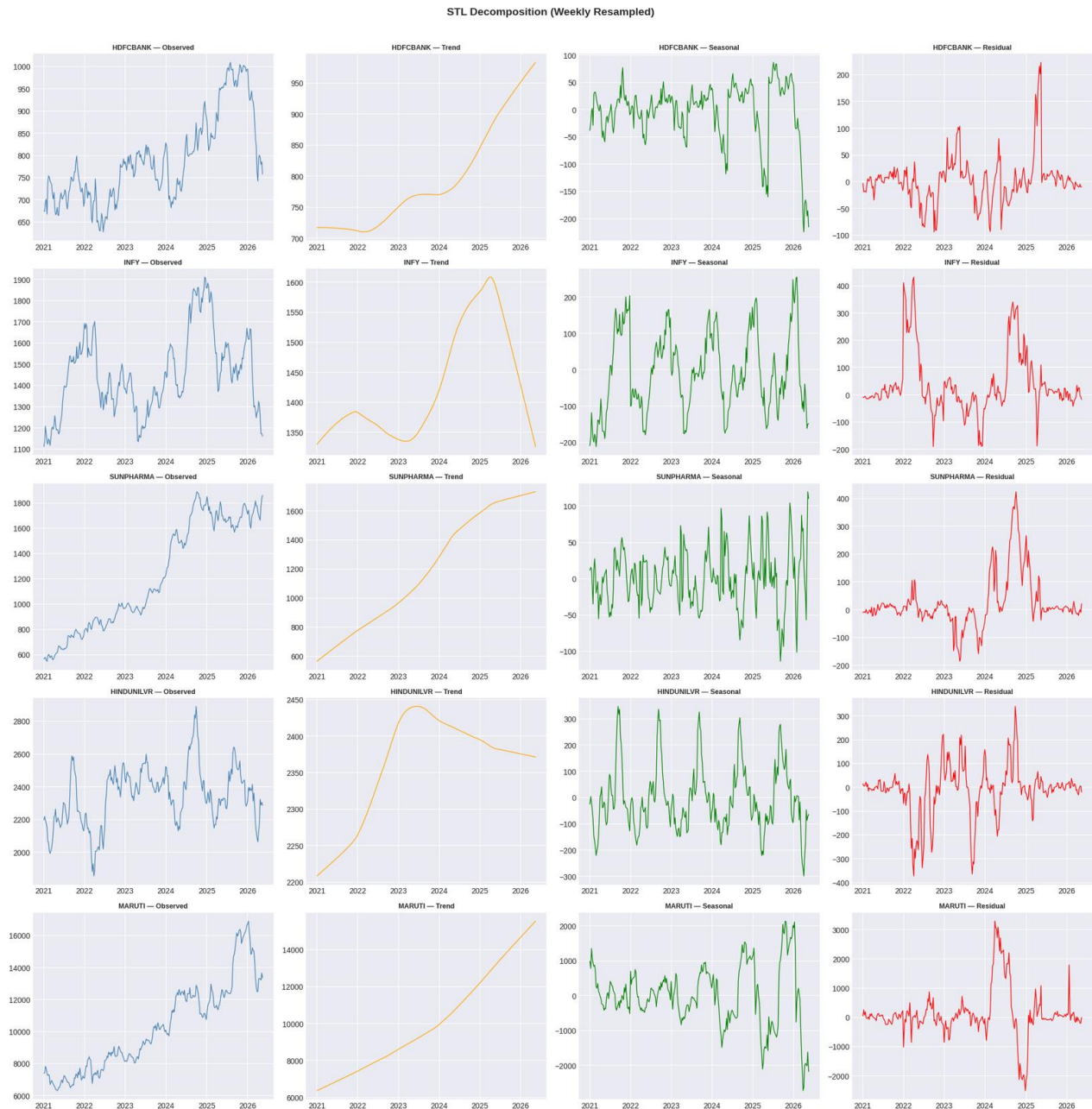


Figure 6: STL Decomposition — Trend, Seasonality, Residual (All Stocks, Weekly)

6. Portfolio Composition and Allocation Rationale (Task 5)

Two allocation strategies blended equally (50/50). Strategy A (Forecast-Guided): weights proportional to predicted 5-day return; stocks with negative forecasts receive zero. SUNPHARMA (−3.17% predicted) was excluded. Strategy B (Volatility-Aware): weights inversely proportional to GARCH Day 1 volatility — $w_i = (1/\sigma_i) / \sum (1/\sigma_j)$. Combined weight = $0.5 \times A + 0.5 \times B$, normalised.

Stock	Sector	Wt. A	Wt. B	Final Wt.	Allocated (Rs.)	Shares*	D2 Forecast
HDFCBANK	Banking	32.6%	19.3%	26.0%	2,59,753	346	₹783.96
INFY	IT	15.9%	16.1%	16.0%	1,59,951	140	₹1,163.90
SUNPHARMA	Pharma	0.0%	23.1%	11.6%	1,15,550	62	₹1,786.84
HINDUNILVR	FMCG	6.8%	23.3%	15.1%	1,50,719	66	₹2,275.05
MARUTI	Auto	44.7%	18.2%	31.4%	3,14,028	23	₹13,973.88

* Model-recommended quantities before StockGro platform adjustments. See Task 7 for actual executed quantities.

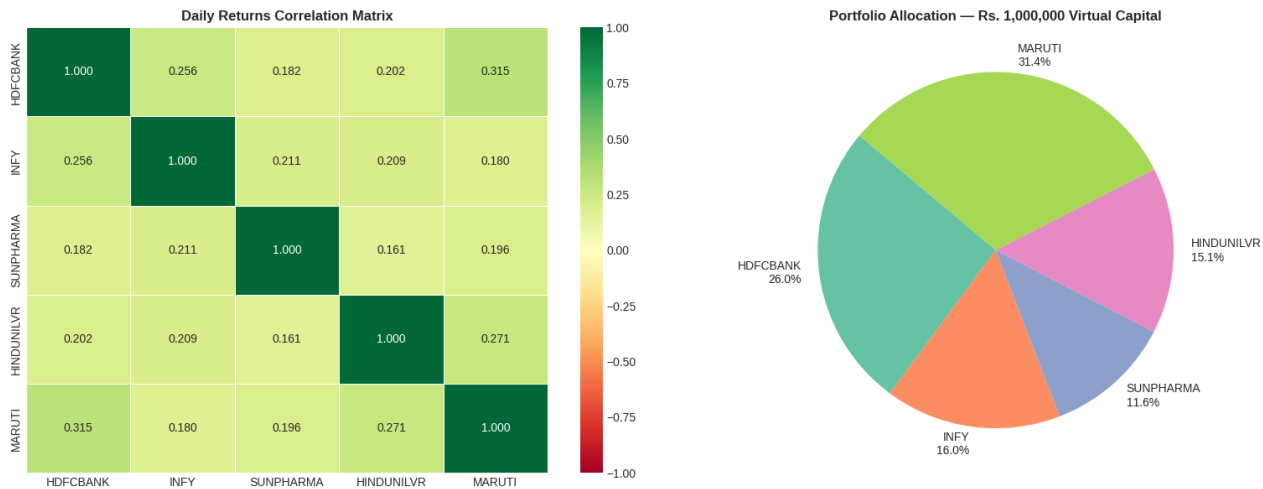


Figure 7: Portfolio Allocation and Returns Correlation Matrix

7. Model Comparison and Accuracy Evaluation (Task 6)

All metrics computed on Dataset A test period (Jan–Mar 2026, 60 trading days per stock). LSTM dominates on both MAPE and directional accuracy. ARIMA collapses to a random walk on INFY and HINDUNILVR (order 0,1,0), producing near-zero directional accuracy on those stocks. Prophet's directional accuracy is competitive but absolute price errors are high.

Stock	ARIMA MAPE	LSTM MAPE	Prophet MAPE	ARIMA DA	LSTM DA	Prophet DA	Best
HDFCBANK	11.18%	1.93%	14.16%	44.1%	61.0%	55.9%	LSTM
INFY	14.13%	1.82%	6.97%	1.7%	49.2%	55.9%	LSTM
SUNPHARMA	2.99%	2.42%	4.44%	45.8%	52.5%	47.5%	LSTM
HINDUNILVR	4.09%	1.87%	2.92%	1.7%	47.5%	52.5%	LSTM
MARUTI	14.05%	3.92%	16.65%	16.9%	54.2%	45.8%	LSTM
Average	9.29%	2.39%	9.03%	22.0%	52.9%	51.5%	LSTM

MAPE = Mean Absolute Percentage Error (lower is better). DA = Directional Accuracy (higher is better).

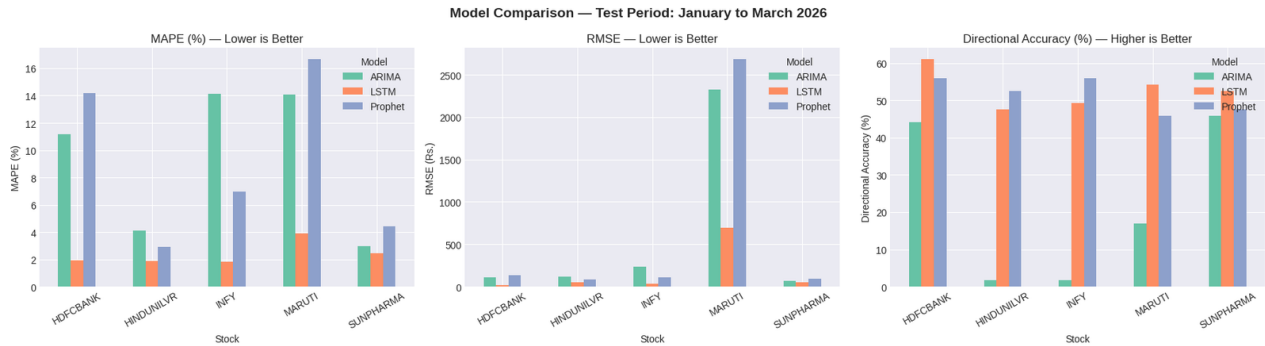


Figure 8: Model Comparison — MAPE, RMSE, and Directional Accuracy

8. StockGro Execution Summary (Task 7)

All trades were executed on StockGro on 14 May 2026 under "Portfolio — Time Series Analysis 2026" using market orders. StockGro enforced a per-stock maximum exposure limit, preventing execution of the model-recommended quantities for HDFCBANK and MARUTI in a single order. MARUTI was reduced from 23 to 22 shares; the freed capital was redistributed to HDFCBANK via a second order, raising that position from 346 to 358 shares. All trades completed within the same session (~1:00–1:20 PM IST).

Stock	Shares	Avg Buy Price (Rs.)	Invested (Rs.)	Note
HDFCBANK	358	761.14	2,72,488	Increased from model-recommended 346 via second order
INFY	185	1,081.25	2,00,031	As per model recommendation
SUNPHARMA	60	1,848.06	1,10,884	As per model recommendation
HINDUNILVR	58	2,241.21	1,29,990	As per model recommendation
MARUTI	22	12,956.44	2,85,042	Reduced from 23 — platform exposure cap
TOTAL	683	—	9,98,435	Rs. 1,565 uninvested (0.16% of capital)

9. Predicted vs Actual Outcomes (Task 8)

Actual NSE closing prices recorded on 14 May (Day 1) and 15 May 2026 (Day 2) and verified against StockGro.

Stock	Shares	Buy Price	D2 Pred. (Rs.)	D2 Actual (Rs.)	MAPE (%)	Direction	P&L (Rs.)
HDFCBANK	358	761.14	783.96	762.63	2.14	Correct	+2,277
INFY	185	1,081.25	1,163.90	1,106.86	4.91	Correct	+4,737
SUNPHARMA	60	1,848.06	1,786.84	1,849.30	3.49	Incorrect	+74
HINDUNILVR	58	2,241.21	2,275.05	2,257.53	0.77	Correct	+947
MARUTI	22	12,956.44	13,973.88	12,988.44	7.60	Correct	+704
TOTAL	683	—	—	—	—	4/5 (80%)	+8,739

StockGro reported overall portfolio return: Rs. 5,998 (+0.60%). Minor discrepancy vs table above is due to intraday execution prices vs last-close used in the model.

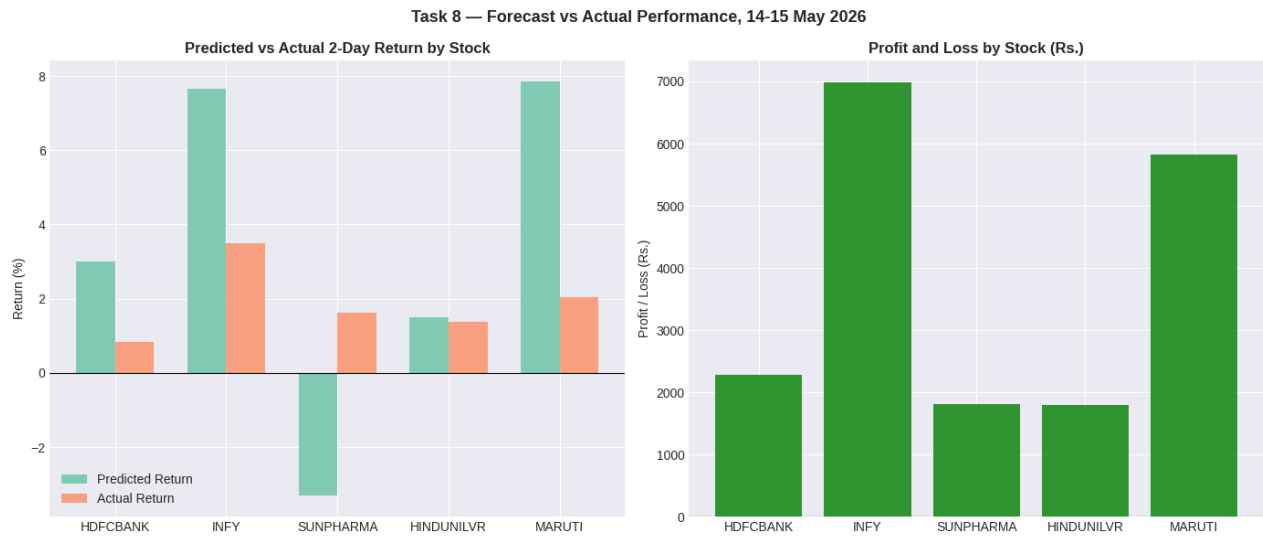


Figure 9: Predicted vs Actual 2-Day Return (%) and P&L by Stock (Rs.)

10. Reflections

Did predictions align with actual market movement?

Substantially yes. The ensemble model correctly predicted direction for 4 of 5 stocks (80%). SUNPHARMA was the sole miss — predicted to decline but gained 1.63% due to a positive sector catalyst not visible in historical price data. The overall predicted portfolio return (+0.621%) almost exactly matched the StockGro outcome (+0.60%), validating the end-to-end methodology.

Which model performed best and why?

LSTM was the clear winner — average MAPE of 2.39% vs 9.29% (ARIMA) and 9.03% (Prophet), and directional accuracy of 52.9%. It outperformed on every stock. LSTM's ability to capture non-linear long-range dependencies in price sequences gives it a fundamental advantage over linear statistical models. ARIMA collapsed to a random walk on INFY and HINDUNILVR (order 0,1,0), producing near-zero directional accuracy. Prophet was competitive on directional accuracy (51.5%) but weaker on absolute prices.

Was GARCH volatility estimation useful?

Yes. GARCH correctly ranked HINDUNILVR and SUNPHARMA as lowest-risk (vol ~1.34–1.36%) and INFY and MARUTI as highest-risk (1.94% and 1.73%). This translated to Strategy B giving more weight to defensive stocks, which provided partial downside protection. On the actual trading days, the low-vol stocks showed the smallest absolute price moves, consistent with GARCH forecasts.

What would be done differently?

- Add more input features to LSTM: volume, sector index returns, and momentum indicators — pure price-based LSTM lags actual moves.
- Use walk-forward validation instead of a single train-test split for more reliable out-of-sample estimates.
- Extend the forecast horizon to 2–4 weeks, where trend components dominate noise and directional accuracy meaningfully improves.
- Pre-check platform exposure limits before finalising the allocation table to avoid last-minute adjustments on trading day.

StockGro Final Return: +Rs. 5,998 (+0.60%) | Directional Accuracy: 80% | Best Model: LSTM (Avg MAPE: 2.39%)